Supplementary material to "Elastic response of aluminium to magnetostriction of an $Al_{13}Fe_4$ inclusion and the extent of possible plasticity"

I. Mikhailovskiy, D. Pshonkin

S1 Building and preparing the cell

The matrix is a periodic aluminium crystal with lattice parameter $a_0 = 4.05$ Å, at which bulk aluminium is stress free with this potential. The cell has 44 periods of 2.864 Å along $x \parallel [1\bar{1}0]$, 25 periods of 4.960 Å along $y \parallel [11\bar{2}]$ and 54 (111) layers of 2.338 Å along $z \parallel [111]$; its size is $126.0 \times 124.0 \times 126.3$ Å. The inclusion is a ball of radius 25 Å cut from an $Al_{13}Fe_4$ crystal with its a axis along x and b axis along y. It contains 4 487 atoms, 3 429 Al and 1 058 Fe (ratio 3.24 against the stoichiometric 3.25). The aluminium atoms inside the ball and all those closer than 2.2 Å to an inclusion atom were removed (4 426 atoms). The shortest distances after construction are 2.20 Å between matrix and inclusion and 2.37 Å inside the inclusion.

With this potential a freely relaxed $Al_{13}Fe_4$ crystal changes its lattice with respect to the measured one: a by +7.5%, b by −2.5%, the cell height by +3.0%, the volume by +7.9%. The inclusion atoms are therefore fixed at the measured lattice at every stage. In a trial run with a free inclusion the cell stretched unevenly along its axes at 300 K, which points to the inclusion swelling inside the aluminium; that run was stopped and is not used.

Preparation of the reference cell: minimisation of the matrix energy to a force two-norm of 1 eV/Å; 10 ps at 300 K (Nosé–Hoover thermostat, matrix atoms only, fixed cell size); cooling to 0.5 K over 10 ps (Langevin thermostat); minimisation in chunks of 2000 iterations to a force two-norm of 3×10^{-4} eV/Å. The mean normal stress of the matrix atoms more than 15 Å from the inclusion surface is then $\sigma_0 = 125.8$ MPa. All cell dimensions are multiplied by $1 - \sigma_0/3K = 0.99946$, where $K = 77.4$ GPa is the bulk modulus of aluminium with the potential (only matrix atoms are moved), and the minimisation is repeated. The mean stress far from the inclusion is then −2.6 MPa.

S2 Convergence and linearity

Each cell is minimised in two stages, to a force norm of 10^{-3} and then 3×10^{-4} eV/Å. Between the stages the stresses in shells more than 5 Å from the inclusion surface change by at most 0.1 MPa.

The $+\varepsilon$ cell starts the minimisation from the elastic solution and the $-\varepsilon$ cell from the atom positions of the reference cell. Half the sum of the responses to $+\varepsilon$ and $-\varepsilon$ therefore measures both nonlinearity and the dependence on the starting point. In shells more than 5 Å from the surface it is at most 0.8 MPa, about 1% of the stresses along and across the field. The main text uses half the difference of the two responses.

No matrix atom jumps to a new position. The number of matrix atoms whose displacement differs from the elastic solution by more than 0.1 Å is 0 in the $+\varepsilon$ cell and 3 in the $-\varepsilon$ cell. All such atoms lie within 3.3 Å of the surface and depart by at most 0.21 Å; no atom departs this much in both cells.

Defects are searched for with DXA (the dislocation extraction algorithm) [1], which finds dislocation lines from the distortion of the lattice around them, and with a nearest-neighbour

analysis, which identifies the crystal surroundings of each atom. The search covers the reference and $\pm\varepsilon$ cells, after each minimisation stage, with and without the inclusion atoms. No dislocation segment is found in any cell. The aluminium atoms without FCC (face-centred cubic) surroundings more than 6 Å from the surface are nearly the same in all three cells: 36–37 atoms 6–7.5 Å from it, at the edge of the interface layer.

The mean added stress over the matrix atoms (xx, yy, zz) is $(-2.8; 1.3; 1.5)$ MPa; the elastic solution for the same cell gives $(-3.7; 1.8; 1.9)$ MPa. Their ratio, about 0.76, is close to the factor 0.80 with which the atomistic field matches the elastic solution (Section 3.2 of the main text).

The atomistic field is weaker than the elastic solution because of a compliant aluminium layer at the interface. The displacements of the matrix atoms, relative to those of the elastic solution for the same cell, are 94% along the surface normal and 86% along the surface in the first layer ($d < 2$ Å), 85% and 73% at $d = 4$–6 Å, and 82% and 72% at $d = 12$–14 Å. The strain of each atom is obtained from the displacements of its neighbours within 3.2 Å (the twelve nearest in FCC). For $d > 8$ Å the virial stress equals the cubic elastic constants of the potential times this strain with a slope of 0.97–0.99 (correlation at least 0.97); for $d < 3$ Å the slope is 0.72–0.80, so the layer at the interface carries less stress for the same strain. The aluminium density at $r = 25.0$–25.5 Å is 0.49 of bulk. The cavity left for the inclusion after removing the aluminium atoms within 2.2 Å of it is 7% larger than the inclusion in volume. The far field alone does not tell such a compliant layer from a perfectly bonded sphere of smaller radius: the factor 0.80 corresponds to a radius of 23.2 Å. The tangential displacement in the first layer is 86% of the elastic one, so at 0 K the interface does not slide freely.

With the inclusion strain 4 and 10 times the given one (0.78 and 1.94%, minimised to 10^{-3} eV/Å) there are no defects either: DXA finds no dislocations, there are no more atoms without FCC surroundings beyond 6 Å from the surface than in the reference cell, and the field divided by the strain multiple equals the field at ε (median ratios 1.01 and 0.99). At 10ε the shear stress in the lobes reaches 306 MPa. The calculation is static and at 0 K, so it does not include thermally activated nucleation of dislocations. Nor does the cell contain pre-existing dislocations, whose motion sets the yield stress of the alloy, so the absence of defects does not mean that the alloy near such an inclusion deforms only elastically.

S3 Given strain and free magnetostriction

The inclusion atoms are fixed, so the strain (1) of the main text is the actual strain of the inclusion inside the matrix. The matrix field depends only on the displacement of the inclusion surface and on the elasticity of the matrix. Consider an elastic spherical particle with shear modulus $k\mu$, where μ is the shear modulus of aluminium, and free magnetostriction λ of the same form as (1), that is, the strain the particle would have in the field without the matrix. Its actual strain inside the matrix is smaller than λ. Eshelby's solution for an inhomogeneous inclusion, one whose elastic constants differ from those of the matrix [2, 3], gives for a deviatoric strain (one that does not change the volume, like (1))

$$\varepsilon = A\lambda, \qquad A = \frac{\beta k}{1 + \beta(k-1)}, \qquad \beta = \frac{2(4-5\nu)}{15(1-\nu)}, \tag{S1}$$

where ν is Poisson's ratio (taken equal for inclusion and matrix for simplicity); for $\nu = 0.344$, $\beta = 0.463$. For a given λ all matrix stresses equal the computed ones multiplied by $A\lambda/\varepsilon$. For an inclusion as stiff as the matrix ($k = 1$) $A = 0.46$; for inclusions 2 and 2.5 times stiffer, $A = 0.63$ and 0.68; for a rigid inclusion $A = 1$. This potential gives the elastic constants of $Al_{13}Fe_4$ at the equilibrium lattice it predicts for this compound, which is overestimated (Section S1), so they are not used to estimate k.

S4 Elastic solutions

Single sphere. A spherical inclusion of radius R with a uniform transformation strain $\boldsymbol{\varepsilon}^*$ (the strain it would take if it were removed from the matrix) in an infinite isotropic matrix has a uniform strain $\boldsymbol{\varepsilon} = \mathbf{S} : \boldsymbol{\varepsilon}^*$ inside, where $\mathbf{S}$ is the Eshelby tensor [2, 3]. To make the interior strain equal the given (1), the transformation strain is set to $\boldsymbol{\varepsilon}^* = \mathbf{S}^{-1} : \boldsymbol{\varepsilon}$; for a deviatoric strain this is $\boldsymbol{\varepsilon}/\beta$. The strain outside, $\varepsilon_{ij}(\mathbf{x}) = D_{ijkl}(\mathbf{x})\, \varepsilon^*_{kl}$, follows from the exterior Eshelby tensor [4]; with $\rho = R/r$ and $\mathbf{n} = \mathbf{x}/r$

$$30(1-\nu)\rho^{-3} D_{ijkl} = (3\rho^2 + 10\nu - 5)\delta_{ij}\delta_{kl} + (3\rho^2 - 10\nu + 5)(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + 15(1-\rho^2)\delta_{ij} n_k n_l$$
$$+15(1-2\nu-\rho^2)\delta_{kl} n_i n_j + 15(\nu-\rho^2)(\delta_{ik} n_j n_l + \delta_{jl} n_i n_k + \delta_{il} n_j n_k + \delta_{jk} n_i n_l) + 15(7\rho^2 - 5) n_i n_j n_k n_l. \tag{S2}$$

The stress is $\sigma_{ij} = \lambda_L \varepsilon_{kk}\delta_{ij} + 2\mu\varepsilon_{ij}$ with the isotropic constants of aluminium for the potential ($\mu = 26.9$ GPa, $\nu = 0.344$, Voigt–Reuss–Hill average of C_{11}, C_{12}, C_{44}). The program was checked on a centre of dilatation ($\boldsymbol{\varepsilon}^* = 10^{-3}\mathbf{I}$): at $r = 1.5R$ it gives $\sigma_{rr} = -21.77$ and $\sigma_{\theta\theta} = 10.89$ MPa, in agreement with Love's exact solution.

The same periodic cell. For a periodic array of inclusions with the cell period the equilibrium equations are solved in Fourier space [5]. For a homogeneous medium of stiffness $\mathbf{C}$ (the cubic constants of aluminium for the potential, rotated to the cell axes) and a transformation strain field $\chi(\mathbf{x})\boldsymbol{\varepsilon}^*$, with χ equal to one in the inclusion,

$$\hat{\varepsilon}_{ij}(\mathbf{k}) = \tfrac{1}{2}\big[\xi_i N_{jm}(\boldsymbol{\xi}) + \xi_j N_{im}(\boldsymbol{\xi})\big]\tau_{ml}\xi_l\, \hat{\chi}(\mathbf{k}), \qquad \mathbf{N} = (\boldsymbol{\xi}\cdot\mathbf{C}\cdot\boldsymbol{\xi})^{-1}, \quad \boldsymbol{\tau} = \mathbf{C} : \boldsymbol{\varepsilon}^*, \tag{S3}$$

with $\boldsymbol{\xi} = \mathbf{k}/|\mathbf{k}|$; the $\mathbf{k} = 0$ term is left out, so the mean strain of the cell is zero, as in the fixed-size calculation. The grid spacing is 1 Å and the inclusion boundary is smoothed over one grid cell. $\boldsymbol{\varepsilon}^*$ is adjusted until the mean strain inside the inclusion equals the given (1): in the periodic cell it has to be 1.07–1.11 times larger, depending on the component, than for the single sphere. The same calculation gives the displacements used as the starting guess for the matrix atoms in the $+\varepsilon$ cell. The mean added stress in the matrix at fixed cell size is known exactly, $\langle\Delta\boldsymbol{\sigma}\rangle = -\mathbf{C} : \boldsymbol{\varepsilon}\, f/(1-f)$; for $f = 3.3\%$ it is $(-3.7; 1.8; 1.9)$ MPa.

S5 The creep model

A dislocation passes an obstacle by a thermal fluctuation at a rate $\omega_0 \exp(-\Delta G/k_\mathrm{B}T)$, where ω_0 is the attempt frequency and ΔG the height of the energy barrier. A shear stress τ does work $V^*\tau$ along the path over the barrier, so it lowers the barrier for forward jumps and raises it for backward ones: $\Delta G_\pm = \Delta G_0 \mp V^*\tau$ [6, 7]. If stresses of both signs occur with equal weights in the volume around the inclusions, the mean rate factor is $\frac{1}{2}[\exp(x) + \exp(-x)] = \cosh x$, $x = V^*\tau/k_\mathrm{B}T$. For dilute inclusions whose fields do not overlap, the relative increase of the mean rate in a material with an inclusion volume fraction f is the integral of $\cosh x - 1$ over the matrix around one inclusion times the number of inclusions per unit volume, f/V_p; this is Eq. (2) of the main text. For a field $\tau(r) = \tau_m(R/r)^3$, where τ_m is the shear stress at the inclusion surface, the substitution $u = x_0(R/r)^3$ reduces the integral to

$$H = f\, x_0 \int_0^{x_0} \frac{\cosh u - 1}{u^2}\, \mathrm{d}u, \qquad x_0 = \frac{V^*\tau_m}{k_\mathrm{B}T}, \tag{S4}$$

which gives the solid curve of Fig. 6. The dashed curve uses the computed field of the single sphere: the integral (2) is taken on a spherical grid (120 radial nodes evenly spaced in $(R/r)^3$, which cover the whole matrix, and 48×48 angular nodes) with the shear stresses on the twelve $\{111\}\langle 110\rangle$ slip systems for 64 random grain orientations. The amplitude τ_m for which Eq. (S4)

gives the same H is then found. A 25% increase is reached at $V^* = 23\,b^3$ (average over grain orientations), $24\,b^3$ (one orientation, that of the cell) and $18\,b^3$ (for the same orientation, the largest of the shear stresses on the twelve slip systems at every point). Since the field is proportional to ε, H depends only on the product $V^*\varepsilon$, and at $V^* = 70\,b^3$ a 25% increase needs $\varepsilon \cdot 23/70 \approx 0.065\%$.

S6 Lattice resistance to dislocation glide

To judge whether the motion of a dislocation past the inclusion can be followed atomistically, the stress at which a straight edge dislocation glides with this potential was measured in a cell without an inclusion. The cell is an aluminium slab of $126.0 \times 19.8 \times 125.5$ Å (18 768 atoms), periodic along x and y with free surfaces along z; the three top and three bottom (111) layers are rigid. An edge dislocation with Burgers vector $b = 2.864$ Å along x and line along y is put into the slip plane at a height of 91.2 Å: the upper half of the slab has one more column of atoms along x than the lower half. The top block is moved along x in steps of 0.00455 Å, and after each step the energy is minimised to a force norm of 10^{-3} eV/Å at 0 K. The applied shear stress is the force of the matrix on the top and bottom blocks divided by the area L_xL_y (the mean of the two blocks, which differ by at most 1 MPa).

The stress rises linearly while the dislocation stays put and drops when it takes a step into the next position of lowest energy. The step is 1.41–1.42 Å long, that is $b/2$; the line positions found from the block displacement agree with those found by DXA [1] to within 0.25 Å, and DXA finds two partial dislocations in every frame: in aluminium an edge dislocation splits into two partials separated by a ribbon where the stacking of the (111) planes is faulted. Before each of the three jumps the stress reaches 9.13, 9.00 and 9.01 MPa, so with this potential a straight edge dislocation at 0 K starts to glide at $\tau_P \approx 9$ MPa (the Peierls stress).

A dislocation above the inclusion. A slab of the same cross-section, $126.0 \times 124.0 \times 125.5$ Å (117 363 atoms), holds an $Al_{13}Fe_4$ sphere of radius 25 Å in its lower half; the slip plane is 12 Å above the top of the sphere, and as built the sphere centre is 15 Å ahead of the line along x. The inclusion is held as in the field calculations, and the blocks are rigid, as in the calculation without an inclusion. The matrix is relaxed by conjugate gradients to a force norm of 10^{-3} eV/Å without annealing, and the top block is then moved along z until the mean σ_{zz} of the matrix is close to zero. During this relaxation the line moves 5.4 Å away from the inclusion and stops 20.4 Å from the sphere centre along x. The line position is the mean position of the two partial dislocations found by DXA; their separation is 9.2 Å in every state. The inclusion atoms are then given the strain $+\varepsilon$ and the matrix is relaxed again: the line moves a further 0.78 Å away from the inclusion. After the inclusion atoms are put back and the matrix is relaxed once more, the line is 0.06 Å from its first position.

Shearing these slabs did not give a separate measure of the effect of the field. Even without load the top and bottom blocks carry shear stresses of 77 and 24 MPa; the held inclusion transmits the difference, so the stress at which the line breaks away is set mainly by the forces that hold the inclusion in place, and the contribution of the inclusion strain cannot be separated in these runs.

S7 Data and programs

Programs and inputs (paths from the repository root):

- the cell: `scripts/build_embedded_inclusion.py`; the strained cells: `scripts/make_strained_inclusion.py`; the LAMMPS inputs: `lammps/embedded_inclusion/in.relax_control` and `lammps/embedded_inclusion/in.static_pair`;

- stresses: `analysis/python/embedded_inclusion_field.py`; convergence and linearity checks: `analysis/python/embedded_gates.py`; defect search: `analysis/python/embedded_dxa.py`; strain, stress per unit strain and density at the interface: `analysis/python/embedded_interface.py`; elastic solutions: `analysis/python/embedded_inclusion_continuum.py`;

- the creep model: `analysis/python/embedded_creep.py`; the yield threshold at the particle: `analysis/python/embedded_yield.py`; elastic constants of the potential: `analysis/python/meam_elastic_constants.py`; figures: `analysis/python/embedded_figures.py`;

- the dislocation slab: `scripts/build_embedded_dislocation.py`; relaxation: `lammps/embedded_inclusion/in.disl_relax`; dislocation positions: `analysis/python/embedded_disl_states.py`; shear: `lammps/embedded_inclusion/in.disl_sweep` and `analysis/python/embedded_sweep.py`;

- the runs of the $\pm\varepsilon$ cells: `scripts/queue_embedded_static.sh`; the runs after them (the dislocation above the inclusion, the cells with strains 4ε and 10ε, shear): `scripts/queue_embedded_plasticity.sh`; records of the dislocation runs and of the 4ε and 10ε cells: `analysis/python/embedded_records.py`.

Result records: `docs/reports/embedded_field.json`, `docs/reports/embedded_gates.json`, `docs/reports/embedded_continuum.json`, `docs/reports/embedded_creep.json`, `docs/reports/embedded_yield.json`, `docs/reports/embedded_peierls.json`, `docs/reports/embedded_dxa.json`, `docs/reports/embedded_interface.json`, `docs/reports/embedded_margin.json`, `docs/reports/embedded_disl_states.json`, `docs/reports/meam_elastic_constants.json`. The configurations and run logs, compressed and with checksums, are in `data/publication/embedded`.

References

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